

Arogya-Nidhi

Health Scheme Eligibility Checker

Android App Development Project Report

Project Title	Android App Development using GenAI – Arogya-Nidhi (Healthcare) – Smart Health Scheme Eligibility Checker
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1. Introduction

Arogya-Nidhi is an Android application developed to help rural families and low-income households in India identify government health schemes they are eligible for, and to guide them through the documentation process required to claim those benefits. The application name combines Sanskrit and Hindi, meaning "Health Wealth" or "Health Treasure" — reflecting its mission to make government healthcare entitlements a genuine and accessible resource for every citizen.

Developed using Kotlin with XML-based layouts, the app integrates a Decision Tree eligibility engine for scheme matching, Room Database for persistent local storage of scheme and hospital data, and a Stepper UI for a guided, empathetic user experience. The application targets families in rural India who are eligible for healthcare schemes such as Ayushman Bharat – PM-JAY, State-specific Free Treatment Cards, and BPL health entitlements, but who are unaware of their eligibility or unprepared with the required documentation.

The application addresses three critical needs of underserved citizens: determining which government health schemes a family qualifies for based on their income, occupation, and BPL status; generating a personalised checklist of documents required for each eligible scheme; and locating the nearest empanelled hospital that accepts those schemes within their district.

Arogya-Nidhi is designed to function entirely offline for its core eligibility and document-guidance features, requiring no internet connection during use. Scheme data and hospital lists are stored persistently on-device using Room Database, ensuring that families in areas with poor connectivity can access the tool reliably. The app targets the vast and underserved population of informal workers, agricultural labourers, and BPL household members across rural India who lack access to healthcare counselling services or government welfare navigation systems.

2. Problem Statement

India's public health infrastructure includes numerous government schemes — Ayushman Bharat PM-JAY, Chief Minister's health funds, state-specific Free Treatment Cards, and Rashtriya Swasthya Bima Yojana — designed to provide financial protection to its most vulnerable citizens. Despite significant government investment, a large proportion of eligible families do not benefit from these schemes. Several systemic problems contribute to this gap:

Challenge	Description
Awareness Gap	Families are unaware of which schemes exist, what they cover, or that they are eligible at all. Information is buried in government portals that require literacy and internet access.
Documentation Confusion	Even families who know about schemes fail to claim benefits because they arrive at hospitals or offices without the correct documents, causing delays, repeated visits, and eventual abandonment of claims.
Language & Literacy Barriers	Government scheme information is predominantly available in English or formal regional language, inaccessible to functionally illiterate users or those with limited formal education.
No Hospital Guidance	Families do not know which nearby hospitals are empanelled under their eligible scheme, leading to rejection at private hospitals or out-of-pocket expenses when free treatment was available.
Out-of-Pocket Expenditure	Without guidance, families pay for surgeries and treatments that should have been covered under government schemes, pushing them into medical debt and financial distress.

These interconnected challenges highlight the urgent need for a dedicated, empathetic, and offline-capable digital counsellor that guides families through eligibility determination and documentation preparation using a simple conversational interface accessible on low-cost Android smartphones.

3. Project Vision

Arogya-Nidhi envisions a world where every Indian family — regardless of literacy level, language, or economic status — can determine their healthcare entitlements in minutes, arrive at a hospital fully prepared, and receive the free or subsidised treatment their family deserves.

The project is built on four foundational pillars:

- **Accessibility** – Healthcare information must reach families in their hands, in their language, on their phones. The app is designed for minimum API 21 devices and requires no internet for its eligibility and document guidance features.
- **Empathy** – The interface must feel like a conversation with a trusted local health worker, not a government form. A Stepper UI with simple questions, visual feedback, and reassuring language ensures that users of all literacy levels can navigate the eligibility quiz independently.
- **Accuracy** – The eligibility engine must correctly map each family's income, BPL status, occupation, and state to the precise set of schemes they qualify for, with no false positives that would lead to wasted effort or false negatives that would cause entitlements to be missed.
- **Actionability** – Knowing eligibility is not enough. Arogya-Nidhi produces a concrete, printable document checklist and a searchable list of empanelled hospitals so that every result leads directly to the next step a family must take.

Arogya-Nidhi is not merely an information app; it is a healthcare navigation tool designed to convert government entitlement into real-world health outcomes for India's most vulnerable citizens.

4. Scope of the Application

4.1 Target Users

- Rural and semi-urban families with household incomes below the threshold for major government health schemes
- BPL (Below Poverty Line) cardholders seeking to understand their healthcare entitlements
- Agricultural labourers, daily wage workers, and unorganised sector employees
- ASHA workers and community health volunteers who guide families through scheme access
- Any citizen preparing to seek treatment at a government or empanelled private hospital

4.2 Platform

- Android smartphones with minimum API Level 21 (Android 5.0 Lollipop and above)
- Optimised for low-end and mid-range devices with limited RAM and storage
- APK size kept minimal to support devices with limited internal storage common in rural areas

4.3 Offline Functionality

- All scheme eligibility data, document checklists, and hospital lists are stored locally using Room Database — no cloud or server required
- The eligibility quiz, scheme result display, and document guide work fully offline
- Hospital search by district uses pre-loaded data, requiring no internet for lookup

4.4 Key Operational Scenarios

- Completing a 5-question eligibility quiz to determine applicable government health schemes
- Viewing a personalised scheme result listing all schemes a family qualifies for
- Reviewing a document checklist for each eligible scheme to prepare for hospital or office visits
- Searching for the nearest empanelled hospital by district that accepts each eligible scheme

5. Objectives

The project is aligned with the following measurable objectives:

- **Provide a guided eligibility determination flow** – Implement a 5-step Stepper UI that collects family income, BPL status, occupation, state, and family size, then maps these inputs to eligible schemes through a Decision Tree logic engine implemented in Kotlin.
- **Deliver accurate and dynamic scheme results** – Ensure that the scheme result screen correctly reflects all schemes a family qualifies for based on their quiz inputs, with the result changing accurately whenever income or BPL status changes.
- **Generate personalised document checklists** – For each eligible scheme, display a complete and accurate list of required documents (Aadhaar, BPL card, income certificate, etc.) so that the family arrives at the hospital or office fully prepared.
- **Provide a searchable empanelled hospital directory** – Allow users to search for hospitals by district that are registered under each of their eligible schemes, using locally stored hospital data to ensure offline access.
- **Ensure permanent, offline data persistence** – All scheme data, document checklists, and hospital records must be stored persistently in Room Database and survive application restarts and device reboots without any internet dependency.
- **Deliver an empathetic and accessible user interface** – Design the quiz and results screens to be usable by individuals with limited literacy, using visual icons, simple language, large touch targets, and a Stepper UI that prevents overwhelming the user with too much information at once.
- **Support scheme data updates via JSON seeding** – Allow scheme, document, and hospital data to be updated by seeding the Room Database from a bundled JSON asset file, enabling future scheme additions without requiring an app redesign.

6. Features

6.1 Core Features

Feature		Description
Eligibility Quiz		A guided 5-step Stepper UI that collects family income level, BPL card status, primary occupation, home state, and family size. Each step presents a single question with clearly labelled options to minimise cognitive load.
Decision Engine	Tree	A Kotlin-based if-else Decision Tree processes quiz responses to identify all government schemes the family qualifies for. The engine evaluates conditions including income thresholds, BPL status, and occupational category against scheme eligibility rules stored in Room Database.
Scheme Screen	Result	Displays a list of all eligible schemes with their name, coverage amount, and a brief description. Results update dynamically whenever the user modifies their quiz answers. Ineligible schemes are not displayed, reducing confusion.
Document Guide		For each eligible scheme, a dedicated Document Checklist screen lists every required document (e.g., Aadhaar card, BPL card, income certificate, caste certificate, bank passbook). Each document is displayed with a simple icon and a brief explanation of why it is needed.
Hospital Finder		A searchable list of government and empanelled private hospitals filterable by District. Each hospital entry shows its name, address, contact number, and the schemes it accepts. Data is pre-loaded from a JSON asset into Room Database at first launch.
Persistent Storage		Room Database (SQLite) stores all scheme definitions, document checklists, and hospital records permanently on-device. Data is retained after app closure, device restart, or extended inactivity without requiring any internet connection.
Scheme View	Detail	Tapping any scheme in the result list opens a full detail screen showing the scheme's official name, annual coverage amount, eligible beneficiaries, how to apply, and the issuing government body.
Quiz Reset & Retake		Users can retake the eligibility quiz at any time to check eligibility for a different family member or updated income status. Previous results are replaced with the new evaluation.

6.2 Future Scope Features

- **Multi-Language Support** – Extend the interface to support Hindi, Kannada, Tamil, Telugu, and Marathi to reach users in their native language.
- **Online Scheme Sync** – Periodically fetch updated scheme eligibility rules and new scheme additions from a government API or Arogya-Nidhi backend, ensuring the app always reflects the latest policy changes.

- **Application Status Tracker** – Allow users to record which schemes they have applied for and track the status of their application, with reminder notifications for pending documents.
- **PDF Card Generation** – Generate a printable PDF summary of the family's eligible schemes and document checklist that can be shared with a health worker or taken to the hospital.
- **ASHA Worker Mode** – A separate app mode for community health workers to run eligibility checks for multiple families and generate bulk guidance reports.
- **Biometric Authentication** – Optionally secure saved family profiles with fingerprint authentication on supported devices.

7. Functional Requirements

The following functional requirements define the core behaviours the system must support:

FR-1: Eligibility Quiz

- The system shall present a Stepper UI with exactly 5 steps, one question per step, without displaying all questions simultaneously.
- Step 1 shall capture annual household income (e.g., Below ₹1 Lakh, ₹1–2.5 Lakh, ₹2.5–5 Lakh, Above ₹5 Lakh).
- Step 2 shall capture BPL card status (Yes / No / Applied, Pending).
- Step 3 shall capture primary occupation (e.g., Agricultural Labourer, Unorganised Worker, Government Employee, Salaried Private, Self-Employed, None).
- Step 4 shall capture home state (drop-down list of Indian states).
- Step 5 shall capture family size (1–2 members, 3–5 members, 6+ members).
- The system shall not allow the user to proceed to the next step without answering the current step's question.

FR-2: Decision Tree & Scheme Matching

- The system shall evaluate quiz responses against eligibility rules for all schemes stored in Room Database.
- The Decision Tree shall correctly identify Ayushman Bharat PM-JAY eligibility for families with BPL status or income below ₹2.5 Lakh.
- The system shall support state-specific schemes by filtering results by the home state entered in Step 4.
- The scheme result must change accurately whenever the user modifies their income or BPL status input and retakes the quiz.

FR-3: Scheme Result Display

- The system shall display a list of all schemes the family qualifies for, showing scheme name, annual coverage amount, and a one-line description.
- The system shall display a clear message if no schemes are found for the entered profile.
- Tapping any scheme shall open its detail view with full eligibility criteria, coverage details, and application process.

FR-4: Document Guide

- The system shall display a personalised document checklist for each eligible scheme accessed from the result screen.
- Each document in the list shall include its name and a brief explanation of its purpose for the scheme application.
- The checklist shall include documents common to most schemes (Aadhaar, BPL card) as well as scheme-specific documents.

FR-5: Hospital Finder

- The system shall provide a searchable list of empanelled hospitals filterable by district name.

- Each hospital record shall include its name, full address, contact phone number, and list of accepted schemes.
- The hospital list shall be pre-seeded from a bundled JSON asset file into Room Database on first app launch.
- The hospital list must be searchable by District without requiring any internet connection.

FR-6: Data Management

- All scheme definitions, document checklists, and hospital records shall be stored in Room Database and persist across sessions.
- The JSON seeding process shall check for existing data before inserting to prevent duplicate records on subsequent launches.
- The system shall allow the app developer to update scheme data by replacing the bundled JSON asset and triggering a re-seed on the next launch.

8. Non-Functional Requirements

Category	Requirement	Acceptance Criterion
Performance	Load Time	App must reach the Home screen within 1000 ms of launch on a mid-range Android device.
Performance	Quiz Completion	Scheme results must be computed and displayed within 500 ms of completing the final quiz step.
Usability	UI Simplicity	Any user must be able to complete the eligibility quiz and view results within 5 taps from the home screen.
Usability	Empathetic Design	All primary text must use a minimum font size of 14sp. Result card values must use a minimum of 22sp.
Reliability	Data Persistence	100% of scheme and hospital records must be retained after app closure, device restart, and 30+ days of inactivity.
Reliability	Crash Rate	Application crash rate must not exceed 1% during standard quiz and result browsing flows.
Accessibility	Touch Targets	All interactive elements must have a minimum touch target size of 48×48dp per Material Design guidelines.
Compatibility	Android Versions	Application must support Android 5.0 (API 21) through Android 14 (API 34).
Accuracy	Eligibility Correctness	The scheme result must change accurately for every valid combination of income level, BPL status, and state inputs.
Offline Support	Core Features	All core features (quiz, results, document guide, hospital search) must function without any network connection.

9. System Architecture

Arogya-Nidhi follows a single-tier, client-only architecture. All data processing, eligibility evaluation, and scheme matching occur entirely on-device. There are no backend servers, cloud databases, or API dependencies required for the application's core operation.

9.1 Technology Stack

Layer	Technology	Purpose
Language	Kotlin	Primary development language; concise, null-safe, fully interoperable with Android SDK.
UI Framework	XML Layouts + AppCompat	Traditional View-based UI with LinearLayout, RecyclerView, and Material components. Stepper UI implemented using a custom ViewPager2 with progress indicators.
Eligibility Engine	Kotlin Decision Tree	Complex if-else Decision Tree logic that evaluates family income, BPL status, occupation, and state against scheme eligibility rules to determine all applicable schemes.
Local Database	Room Database (SQLite)	ACID-compliant ORM over SQLite. Stores scheme definitions, document checklists, and hospital records permanently on-device with DAO-based queries and migration support.
Data Seeding	JSON Asset + Gson	Scheme, document, and hospital data are bundled as a JSON asset file. On first launch, Gson parses this file and seeds the Room Database using an Executors background thread.
Session Storage	SharedPreferences	Stores user's most recent quiz answers and the first-launch seed completion flag as lightweight key-value pairs.
Build System	Gradle (Groovy DSL)	Dependency management, build configuration, and APK generation including kotlin-kapt for Room annotation processing.
Min. SDK	API Level 21	Targets approximately 98% of active Android devices globally, including low-cost smartphones prevalent in rural India.

9.2 Architectural Overview

The application follows a simplified MVC (Model-View-Controller) pattern appropriate for a single-activity, XML-layout Android app:

- **Model Layer** – The data package contains the Room @Entity classes (Scheme, DocumentItem, Hospital), @Dao interfaces (SchemeDao, HospitalDao), and @Database singleton (AppDatabase). This layer owns all data access logic, JSON seeding, and eligibility rule storage.
- **View Layer** – XML layout files define the UI structure (activity_main.xml, activity_quiz.xml, activity_results.xml, activity_scheme_detail.xml, activity_document_guide.xml, activity_hospital_finder.xml, list_item_scheme.xml, list_item_hospital.xml, list_item_document.xml). Custom drawables provide visual styling for scheme cards, step indicators, and result highlights.
- **Controller Layer** – Activity classes (MainActivity, QuizActivity, ResultsActivity, SchemeDetailActivity, DocumentGuideActivity, HospitalFinderActivity) orchestrate user interaction, invoke the Decision Tree engine, perform database queries on background threads via Executors, and update the UI on the main thread via runOnUiThread.
- **Eligibility Engine** – EligibilityEngine.kt is a standalone Kotlin object that receives the QuizResponse data class as input and returns a List<Scheme> of matched schemes. This separation from the Activity layer makes the engine independently testable.

9.3 Database Design

The Room database contains three tables — schemes, document_items, and hospitals:

Schemes Table:

Column	Type	Constraint	Description
id	INTEGER	PRIMARY KEY, AUTO	Auto-generated unique identifier for each scheme.
schemeName	TEXT	NOT NULL	Official name of the government health scheme.
coverageAmount	TEXT	NOT NULL	Annual coverage or benefit amount (e.g., ₹5,00,000 per family).
description	TEXT	NOT NULL	Brief description of scheme benefits and target population.
minIncome	REAL	DEFAULT 0	Minimum income threshold for eligibility (0 = no minimum).
maxIncome	REAL	NOT NULL	Maximum annual income for eligibility in Indian Rupees.
requiresBpl	INTEGER	DEFAULT 0	1 if BPL card is mandatory for this scheme, 0 otherwise.
stateCode	TEXT	DEFAULT 'ALL'	State code if scheme is state-specific; 'ALL' for central schemes.

10. User Flow

The application is designed to minimise friction and maximise clarity for users who may have limited digital literacy. The complete user journey from app launch to hospital identification is described below:

#	Screen	User Action & System Response
1	Home Screen	App launches. User sees the Arogya-Nidhi home screen with a prominent 'Check Eligibility' button, a brief tagline explaining the app's purpose, and optional access to the Hospital Finder.
2	Quiz – Step 1	User selects their annual household income range from 4 options. A progress bar at the top shows Step 1 of 5. The 'Next' button activates only after a selection is made.
3	Quiz – Step 2	User selects BPL card status (Yes / No / Applied, Pending). Clear icons accompany each option to assist low-literacy users. 'Back' returns to Step 1.
4	Quiz – Step 3	User selects their primary occupation from a list of common informal sector categories. A 'None of the above' option is available.
5	Quiz – Step 4	User selects their home state from a scrollable alphabetical list of Indian states.
6	Quiz – Step 5	User selects family size. Tapping 'See My Results' triggers the Decision Tree engine. The progress bar shows 5/5 complete.
7	Scheme Results	Results screen displays all eligible schemes as cards with scheme name, coverage amount, and a 'View Details' button. A reassuring message is shown at the top (e.g., "Great news! You may be eligible for 3 schemes.").
8	Scheme Detail	Tapping 'View Details' opens a full detail view showing scheme benefits, eligibility criteria, issuing authority, and a 'See Documents Needed' button.
9	Document Guide	Document checklist screen displays all required documents for the selected scheme with icons and brief explanations. User can check off documents they already have.
10	Hospital Finder	User enters their district name in the search bar. The hospital list filters in real-time showing empanelled hospitals, their addresses, contact numbers, and accepted schemes.

11. Technical Implementation

11.1 Stepper UI with ViewPager2

The eligibility quiz is implemented as a 5-step Stepper UI using ViewPager2 with swipe disabled (`setUserInputEnabled(false)`) to enforce linear navigation. Each step is a separate Fragment (Step1Fragment through Step5Fragment) hosted within QuizActivity. A custom horizontal progress bar (using a LinearLayout with dynamically coloured views) provides visual step-completion feedback at the top of the screen. Navigation between steps is handled through a shared QuizViewModel that stores the current step number and the accumulated quiz responses, ensuring that answers are preserved if the user navigates back to a previous step.

11.2 Decision Tree Eligibility Engine

The eligibility engine is implemented as EligibilityEngine.kt, a Kotlin object with a single public function: `evaluate(response: QuizResponse): List<Scheme>`. The function queries all schemes from Room Database via SchemeDao, then applies a multi-condition filter for each scheme: income range check (`response.incomeValue` in `scheme.minIncome..scheme.maxIncome`), BPL status check (if `scheme.requiresBpl == 1`, `response.hasBplCard` must be true), state check (`scheme.stateCode == 'ALL'` or `scheme.stateCode == response.stateCode`), and optional occupation-based conditions for schemes targeting specific worker categories. This approach makes it straightforward to add new schemes and eligibility conditions by updating the Room Database seed data without modifying the engine logic.

11.3 Room Database & JSON Seeding

Scheme, document checklist, and hospital data are stored using the Room persistence library. The AppDatabase singleton uses a double-checked locking pattern to ensure a single instance throughout the app lifecycle. On first launch, a DatabaseSeeder class reads schemes.json, documents.json, and hospitals.json from the assets folder using `context.assets.open()`, parses them with Gson into `List<Scheme>`, `List<DocumentItem>`, and `List<Hospital>` respectively, and inserts them into Room Database using `Executors.newSingleThreadExecutor()`. A SharedPreferences flag (`isDatabaseSeeded`) prevents re-seeding on subsequent launches.

11.4 Background Threading

Room does not permit database operations on the main (UI) thread. All database queries — including eligibility evaluation, hospital list retrieval, and document checklist loading — are executed using `Executors.newSingleThreadExecutor()`. Results are posted back to the UI thread via `runOnUiThread{}`, which updates RecyclerView adapters and result TextViews. This pattern avoids the need for LiveData or coroutines while maintaining thread safety and compatibility with lower API levels.

11.5 Hospital Search Implementation

The Hospital Finder uses a Room DAO query with a LIKE operator to filter hospital records by district: `@Query("SELECT * FROM hospitals WHERE district LIKE '%' || :district || '%")`. The HospitalFinderActivity implements a `SearchView.OnQueryTextListener` that triggers a new database query on every keystroke change (`onQueryTextChange`), delivering real-time filtering without internet access. Results are displayed in a RecyclerView with HospitalAdapter inflating `list_item_hospital.xml` for each entry showing the hospital name, address, phone number, and scheme tags.

11.6 Custom RecyclerView Adapters

Three custom RecyclerView adapters manage the main data views: SchemeAdapter (result cards with scheme name, coverage chip, and detail button), DocumentAdapter (checklist items with icon, name, description, and a Checkbox for the user to mark documents they have gathered), and HospitalAdapter (hospital cards with name, address, phone, and accepted scheme tags). Each adapter uses the DiffUtil pattern to efficiently update only changed items without full list redraws.

11.7 SharedPreferences for Quiz State

The user's most recent quiz answers are saved to SharedPreferences using a named file (QuizPrefs) so that the Results screen can be re-opened without re-running the quiz. When the user opens the app after a previous session, the Home screen offers a 'View Last Results' option if QuizPrefs contains a completed quiz response. This supports users who need to reference their eligibility results multiple times during a hospital visit without re-entering their information.

12. Project File Structure

File / Directory	Purpose
app/build.gradle	App-level Gradle config: plugins, SDK versions, Room, Gson, ViewPager2, and kapt dependencies.
data/Scheme.kt	Room @Entity data class representing a single government health scheme.
data/DocumentItem.kt	Room @Entity data class representing a single required document for a scheme.
data/Hospital.kt	Room @Entity data class representing an empanelled hospital with district and accepted schemes.
data/SchemeDao.kt	Room @Dao interface with queries for scheme retrieval by income range, BPL status, and state code.
data/HospitalDao.kt	Room @Dao interface with district LIKE search query and insert method.
data/AppDatabase.kt	Room @Database singleton with version management, DAO accessors, and migration definitions.
engine/EligibilityEngine.kt	Standalone Kotlin object implementing the Decision Tree evaluation logic. Accepts QuizResponse, returns List<Scheme>.
engine/QuizResponse.kt	Data class encapsulating all five quiz answers: incomeValue, hasBplCard, occupation, stateCode, familySize.
MainActivity.kt	Home screen: Start Quiz button, View Last Results option, and direct link to Hospital Finder.
QuizActivity.kt	Hosts ViewPager2 with 5 step fragments. Manages navigation, progress bar, and final result trigger.
Step1Fragment–Step5Fragment.kt	Individual quiz step fragments: one per question. Each saves its answer to shared QuizViewModel.
ResultsActivity.kt	Displays eligible schemes in a RecyclerView. Loads results from EligibilityEngine on a background thread.
SchemeDetailActivity.kt	Full scheme detail view: name, coverage, description, eligibility, and 'See Documents Needed' button.
DocumentGuideActivity.kt	Document checklist for a selected scheme. Shows icon, name, description, and checkable state per document.
HospitalFinderActivity.kt	Hospital search screen with real-time district filtering using SearchView and Room LIKE query.

DatabaseSeeder.kt	Reads schemes.json, documents.json, hospitals.json from assets and seeds Room Database on first launch.
assets/schemes.json	JSON array of all government health schemes with eligibility rules, coverage amounts, and descriptions.
assets/documents.json	JSON array of document checklist items mapped to scheme IDs.
assets/hospitals.json	JSON array of empanelled hospitals with district, address, phone, and accepted scheme IDs.
res/layout/*.xml	Layout files for all activities, fragments, and list item rows.
AndroidManifest.xml	App manifest: permissions, activity declarations, and launcher activity designation.

13. Permissions & Privacy

Permission	Required For	Notes
No special permissions required	Core features (quiz, results, hospital search)	All core features operate without any dangerous permissions. No network, camera, location, or storage permissions are needed.
READ_EXTERNAL_STORAGE (optional)	Future: PDF export feature	Not used in v1.0. Planned for the PDF document checklist export feature in a future version.

No personal user data is collected, stored externally, or transmitted to any server. All family profile information entered during the quiz is stored exclusively in SharedPreferences on the user's device. Room Database contains only pre-seeded government scheme data and hospital records. The application does not collect analytics, usage statistics, or crash reports. No network permissions are requested, making it technically impossible for the app to transmit any user data. Arogya-Nidhi is designed from the ground up to be a privacy-safe tool. Families can enter sensitive information such as BPL card status and income level with confidence that this data never leaves their device.

14. Impact Goals

14.1 Universal Health Coverage

India's Ayushman Bharat PM-JAY scheme covers over 500 million citizens with up to ₹5 lakh annual hospitalisation coverage, yet a significant proportion of eligible families have never claimed a benefit. Arogya-Nidhi directly addresses the awareness and preparedness gap by making eligibility determination instant and document preparation actionable. Every family that discovers and claims their entitlement represents a real reduction in out-of-pocket medical expenditure that could otherwise push them into poverty.

14.2 Social Justice: Removing the Information Barrier

Access to government welfare information in India is disproportionately concentrated among urban, educated, and digitally connected populations. Rural families, daily wage workers, and agricultural labourers — the primary intended beneficiaries of health schemes — are systematically excluded by the complexity and inaccessibility of government portals. Arogya-Nidhi eliminates this information barrier by delivering a conversational, icon-supported, offline-capable eligibility counsellor directly to the ₹3,000 smartphones in their pockets.

14.3 Financial Protection from Medical Debt

Medical expenses are the leading cause of household financial distress and debt in rural India. Families who pay out-of-pocket for surgeries that should have been covered under government schemes often borrow from informal moneylenders at exploitative interest rates. By ensuring that families know their coverage before a health emergency, Arogya-Nidhi contributes directly to household financial resilience and reduces dependence on predatory credit.

14.4 Empowerment of Community Health Workers

ASHA (Accredited Social Health Activist) workers are the frontline of India's community healthcare network, but they often lack digital tools to guide families through scheme navigation efficiently. Arogya-Nidhi gives every ASHA worker a reliable, offline-capable pocket counsellor that enables faster and more accurate scheme guidance during household visits, multiplying the app's impact beyond individual users to entire communities.

14.5 Scalability of Impact

The JSON-seeded, database-driven architecture of Arogya-Nidhi means that new schemes can be added and eligibility rules updated without requiring a full app redesign. As the Government of India introduces new health policies or modifies existing schemes, the scheme data in the app can be updated by replacing the JSON asset files and triggering a re-seed, ensuring that the app remains accurate and relevant across political cycles and policy changes.

15. Success Criteria

The project will be considered successfully completed when the following criteria are met:

#	Success Criterion	Measurement	Priority
1	The scheme result changes accurately for every valid combination of income level and BPL status input.	Manual functional testing with all input combinations	Critical
2	The Hospital List is searchable by District and returns correct results offline.	Airplane Mode test on device with district queries	Critical
3	The UI is empathetic and simple — any user can complete the quiz within 5 taps.	Manual UI walkthrough with untrained test user	Critical
4	All scheme and hospital records are retained after device reboot and 30+ days of inactivity.	Room DB inspection after reboot test	Critical
5	Document checklist displayed is accurate and complete for each eligible scheme.	Manual validation against official scheme document requirements	Critical
6	Ayushman Bharat PM-JAY is correctly identified for families with BPL card or income below ₹2.5 Lakh.	Functional test with BPL = Yes, Income < 2.5L input	Critical
7	State-specific schemes are displayed only for the correct home state.	Test with two different state inputs for same income profile	High
8	App loads and reaches the Home screen within 1000 ms on a reference device.	Android Profiler measurement	High
9	App does not crash during a complete walkthrough of all screens and features.	Manual regression testing on 3 device tiers	High
10	All features work fully in Airplane Mode.	Airplane Mode test on device for all core flows	Medium

Conclusion

Arogya-Nidhi represents a focused and socially meaningful application of Android development in the domain of healthcare accessibility and government welfare navigation. By addressing a clear and well-defined problem — the inability of India's most vulnerable families to access healthcare entitlements they are legally entitled to — the project delivers measurable practical value alongside significant technical merit.

The application's architecture prioritises accuracy, empathy, and offline-first operation above all else. Every design decision — from the Stepper UI that prevents cognitive overload, to the Decision Tree engine that evaluates eligibility with precision, to Room Database that stores scheme and hospital data reliably without internet — is motivated by a single question: will this work for a family standing at a hospital registration desk in a rural district, with a basic smartphone and no data connection?

The project also establishes a strong technical foundation for future expansion. The JSON-seeded database architecture supports rapid scheme updates without codebase changes. The separation of the eligibility engine from the UI layer makes the Decision Tree independently testable and extensible. Future versions can add multi-language support, online scheme synchronisation, PDF checklist generation, and ASHA worker mode without requiring a fundamental redesign of the existing system.

In conclusion, Arogya-Nidhi is not simply a technical deliverable — it is a small but meaningful step toward universal health coverage and social justice for the millions of Indian families who have remained on the wrong side of the healthcare information divide. Every family that uses this app to discover and claim their entitlement represents a direct, measurable improvement in health outcomes and financial security.